# Project Report

**Topic: -**

**Visualization tool for electric vehicle charge and range analysis using Tableau**

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# 1. Abstract

The rapid adoption of electric vehicles (EVs) worldwide has created a need to analyze EV performance, pricing, infrastructure, and market trends. This project focuses on analyzing multiple EV datasets and presenting insights through data visualization. Using data cleaning techniques and visualization tools, the project identifies patterns in EV pricing, efficiency, charging infrastructure distribution, and brand performance. The results are presented through interactive dashboards and charts to make the insights easy to understand for users.

# 2. Introduction

Electric vehicles are becoming an essential part of the global transportation ecosystem. Governments and automobile companies are investing heavily in EV technology to reduce carbon emissions and promote sustainable transportation. However, analyzing EV data can be complex because it involves multiple variables such as vehicle range, price, efficiency, body style, and charging infrastructure. This project aims to simplify EV data analysis through visualization techniques and interactive dashboards.

# 3. Problem Statement

Although the EV market is growing rapidly, users and analysts often face difficulty understanding trends in EV models, brand performance, and infrastructure availability. The problem addressed in this project is how to collect, clean, analyze, and visualize EV data effectively so that insights about pricing, efficiency, and infrastructure distribution can be clearly understood.

# 4. Objectives

The primary objectives of this project are:

- To collect and analyze electric vehicle datasets.
- To clean and preprocess the data for accurate analysis.
- To visualize EV performance, pricing, and infrastructure data.
- To compare EV brands based on efficiency, speed, and number of models.
- To create an interactive dashboard for easy exploration of insights.

# 5. Data Collection

The datasets used in this project were collected from publicly available EV data sources. The data was downloaded
in CSV format and includes multiple categories of EV information.

Datasets used:

1. EV India Dataset – Contains electric vehicle models available in India with price ranges and vehicle types.
2. Electric Car Specifications Dataset – Includes brand, range, efficiency, top speed, and body style.
3. EV Price Dataset – Contains price information for EV models in different markets.
4. Charging Stations Dataset – Includes location data such as latitude, longitude, and region for EV charging stations.

These datasets provide a comprehensive view of the EV ecosystem including vehicles and infrastructure.

# 6. Data Cleaning and Preparation

Before performing analysis, the raw datasets were cleaned and transformed. Data cleaning is important because raw datasets often contain missing values, inconsistent formats, and unwanted characters.

The following steps were performed during data cleaning:

- Handling missing values (NA values).
- Removing special characters from price columns.
- Converting price and numerical columns to numeric data types.
- Standardizing column names for easier analysis.
- Verifying latitude and longitude values for map visualization.

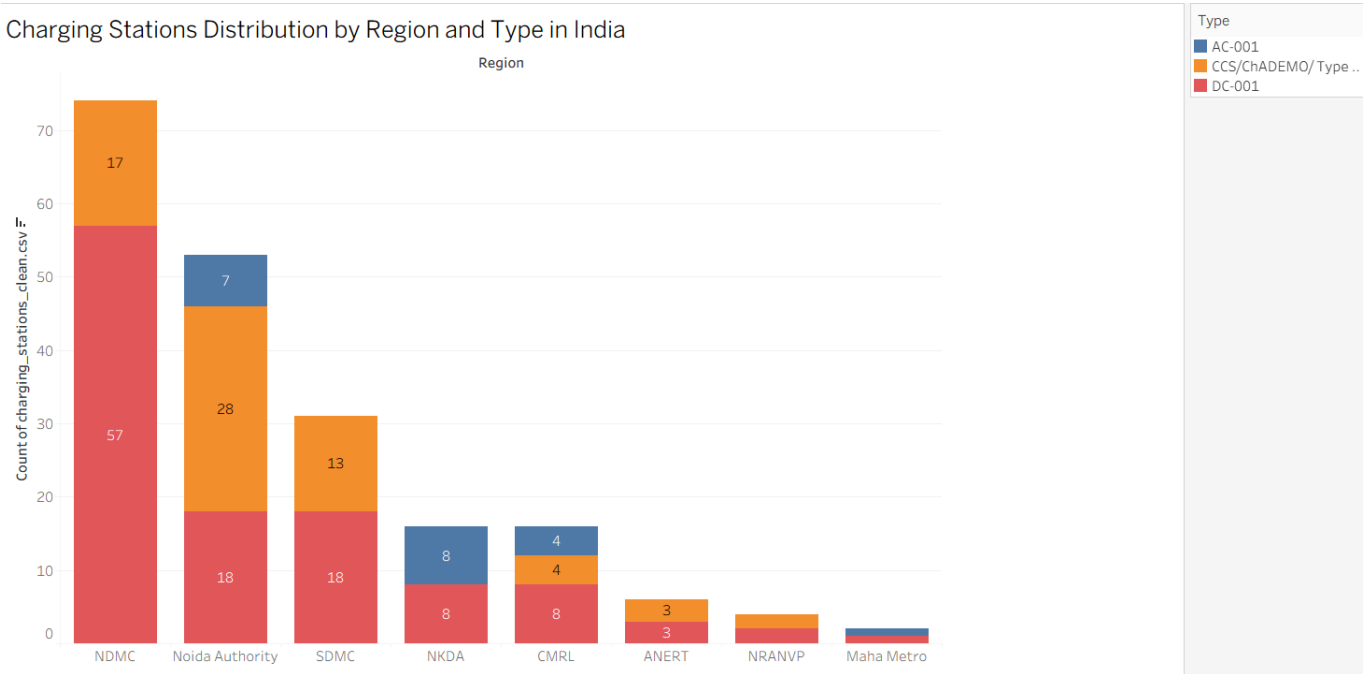
After cleaning, the datasets were ready for analysis and visualization.

# 7. Data Visualization

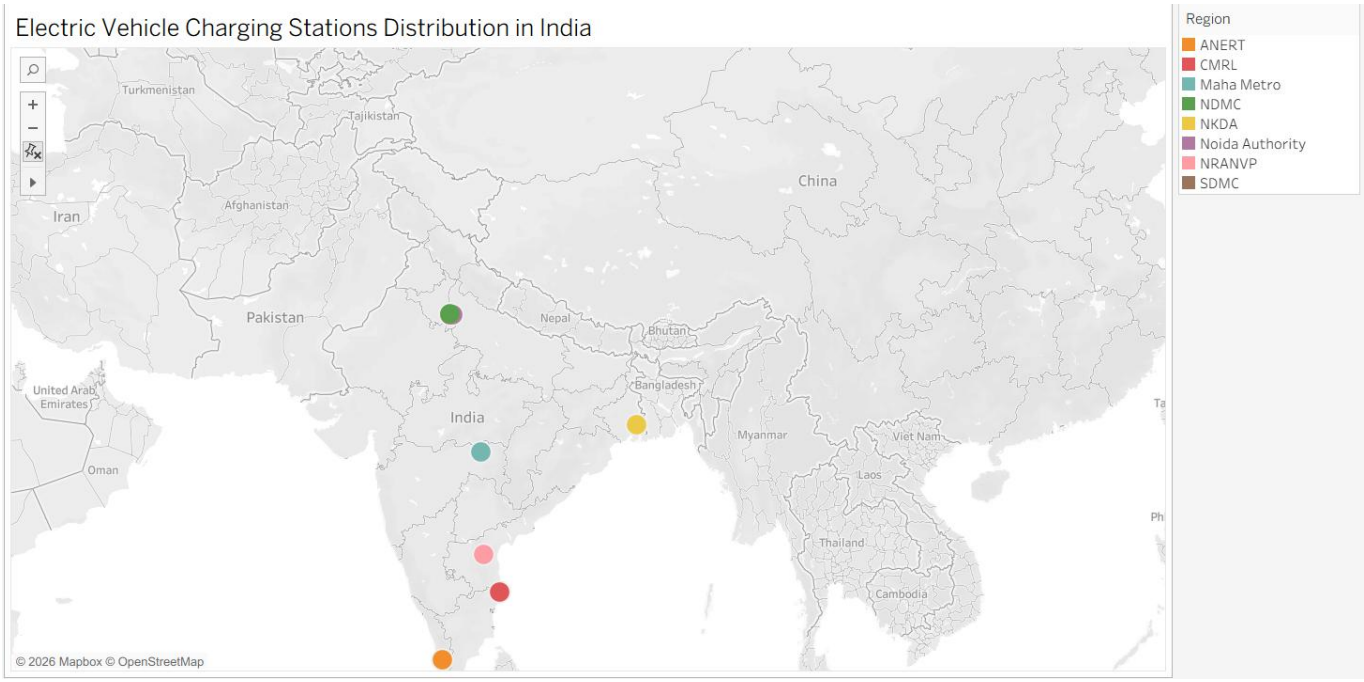
Data visualization plays an important role in understanding patterns and trends in EV data. Several charts were created using Tableau Public to represent different aspects of the EV ecosystem.

Visualizations created:

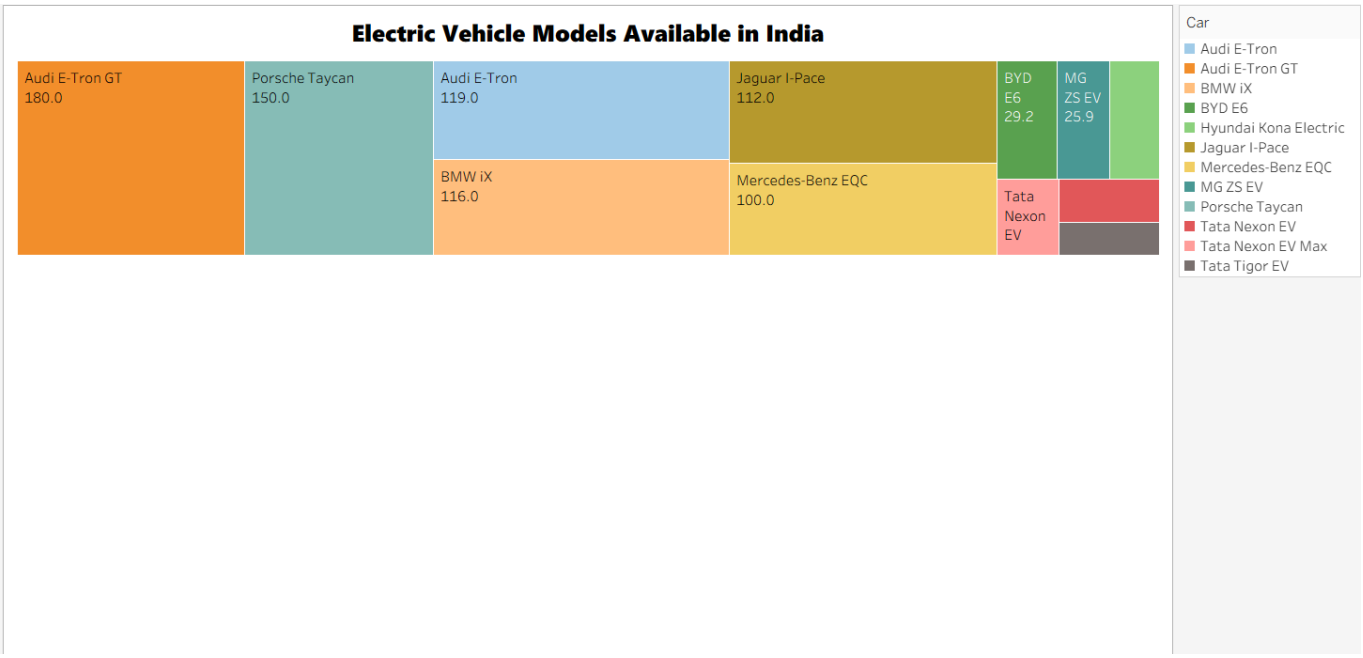
1. Charging Stations by Region – Stacked bar chart showing station distribution.



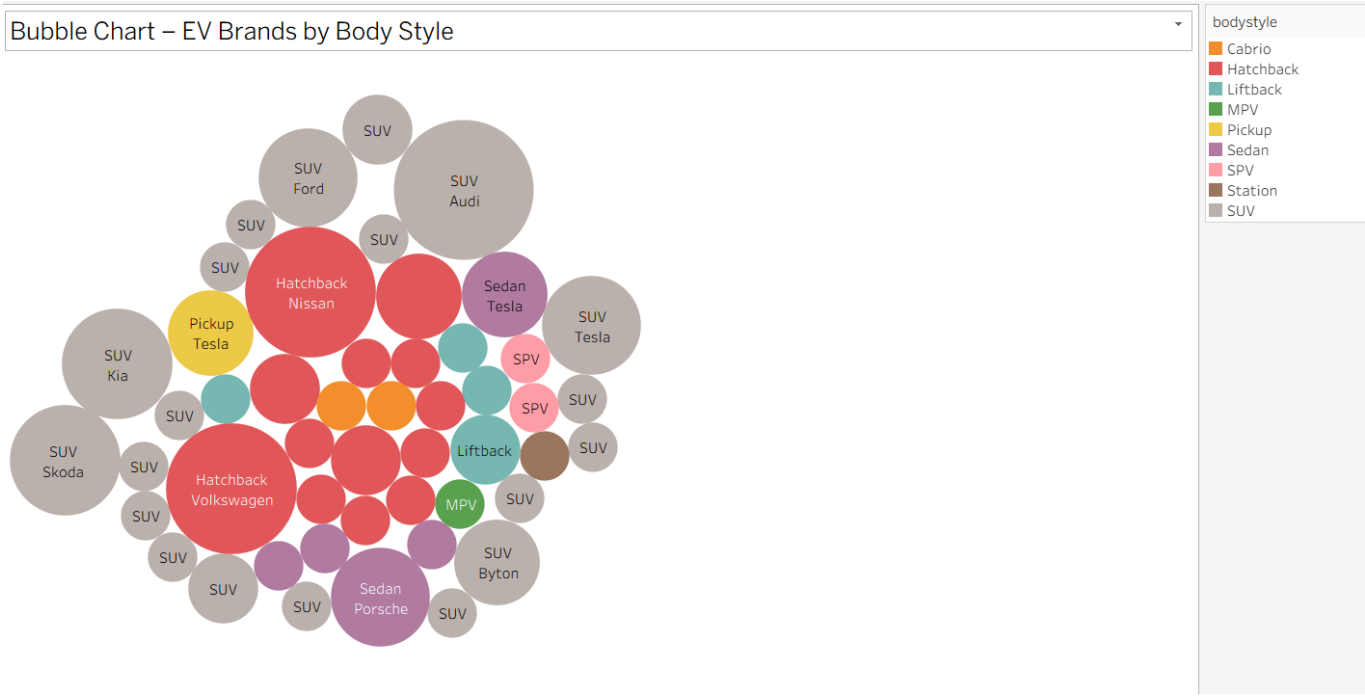
2. Charging Stations Map – Geographic visualization of charging locations.



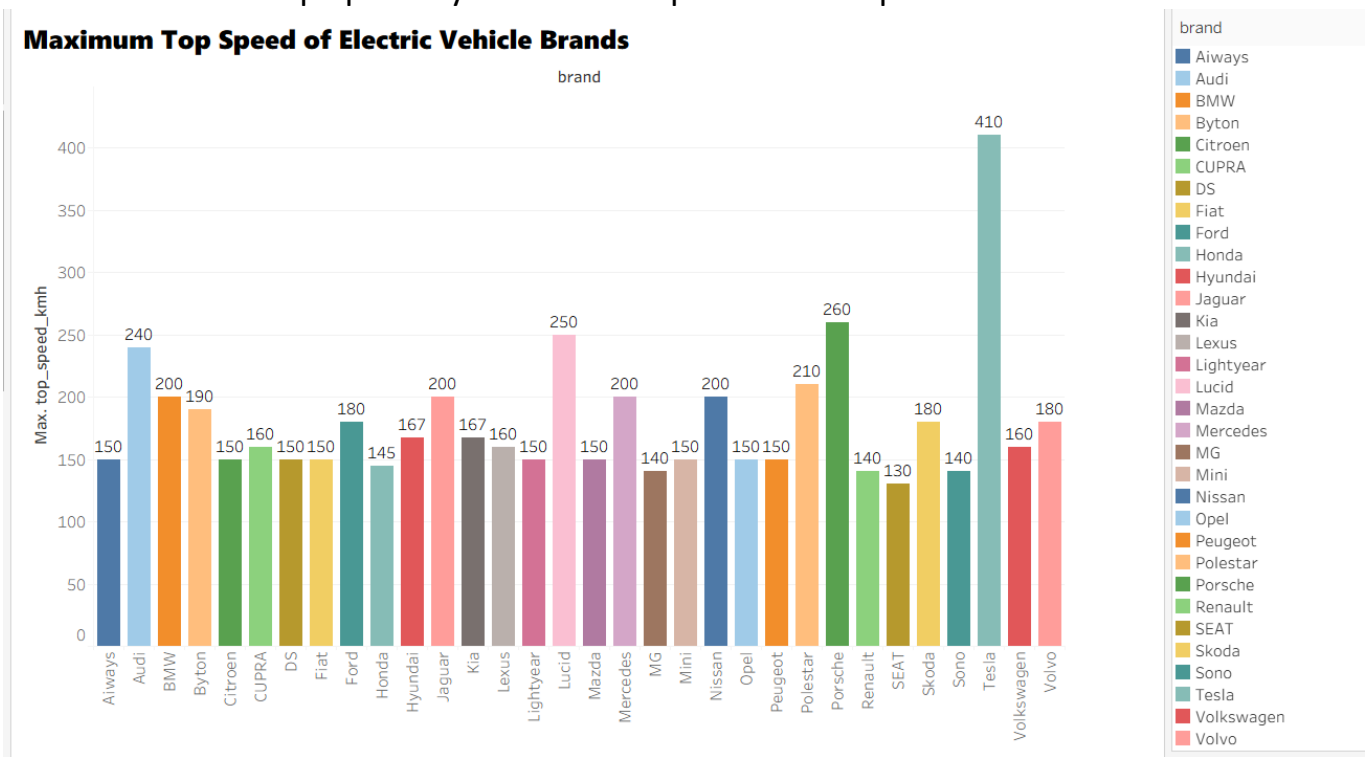

3. EV Models Treemap – Distribution of EV models in the Indian market.



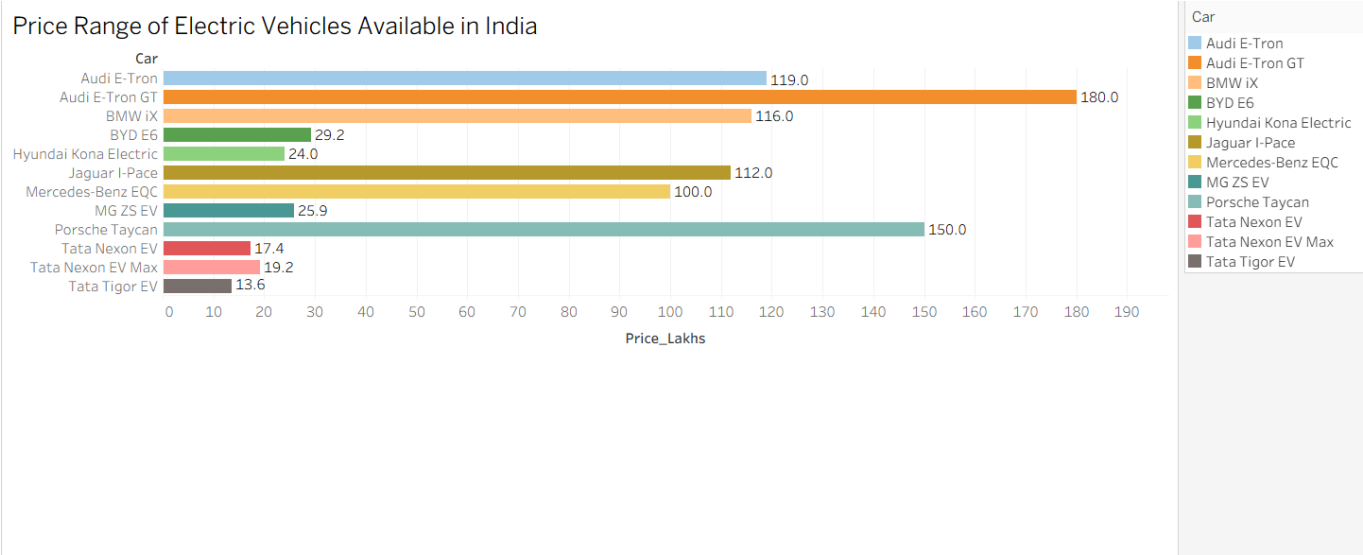
4. EV Brands by Body Style – Bubble chart representing vehicle categories.



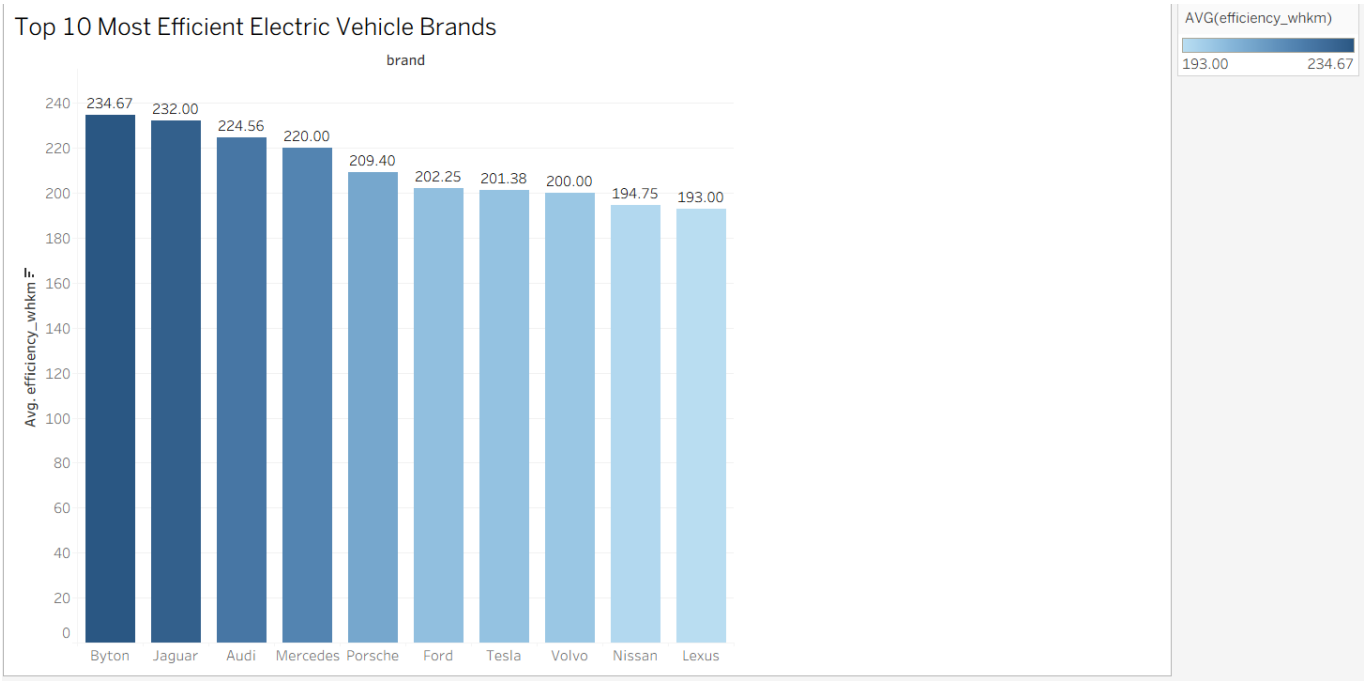

5. Maximum Top Speed by Brand – Comparison of EV performance.



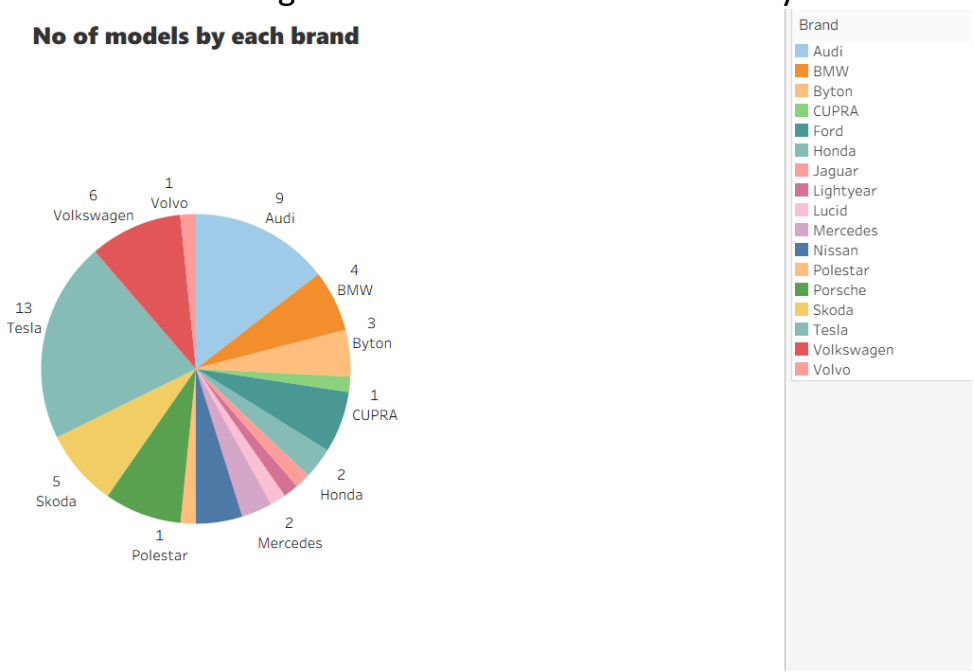
6. EV Price Range in India – Bar chart showing price differences among EV models.




7. Top 10 Efficient EV Brands – Comparison based on energy efficiency.



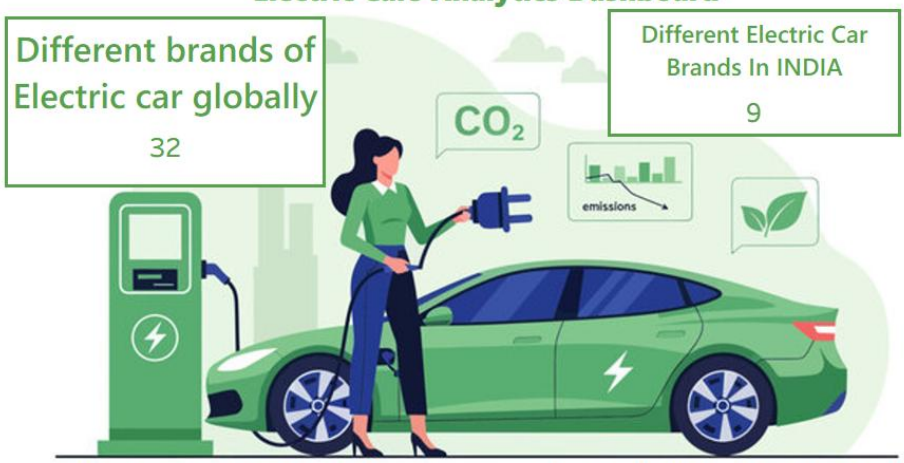
8. EV Models by Brand – Pie chart showing number of EV models offered by brands.



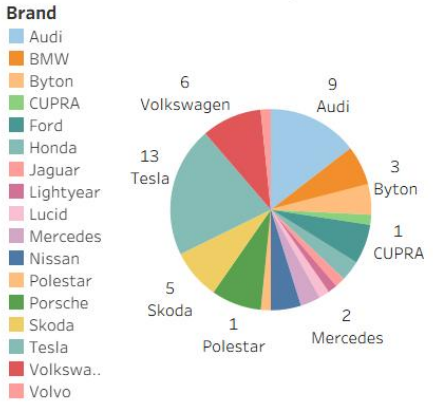

# 8. **Dashboard Creation**

The individual charts were combined into an interactive dashboard using Tableau Public. Dashboards allow users to view multiple insights in one place and interact with the data through filters and highlights. The EV dashboard includes charts for pricing, efficiency, infrastructure distribution, and brand comparison. Users can easily explore different aspects of EV data and identify trends visually.

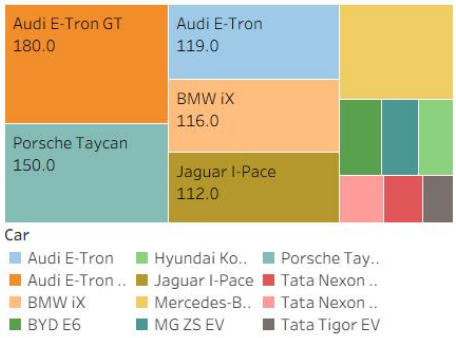
## Electric Cars Analytics Dashboard




### No of models by each brand




### Electric Vehicle Models Available in India





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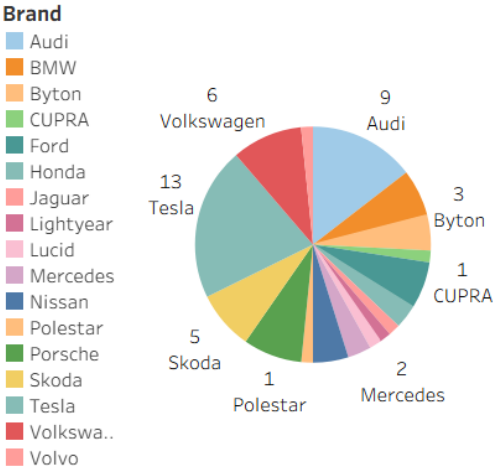


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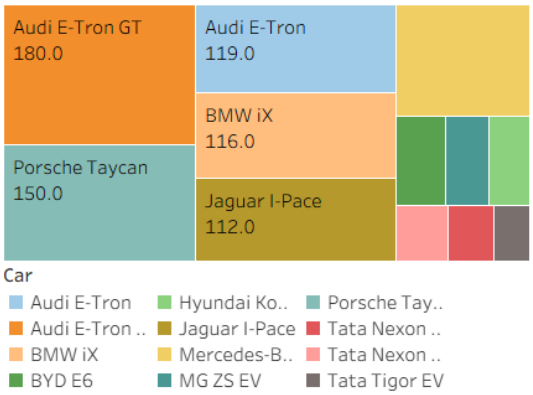
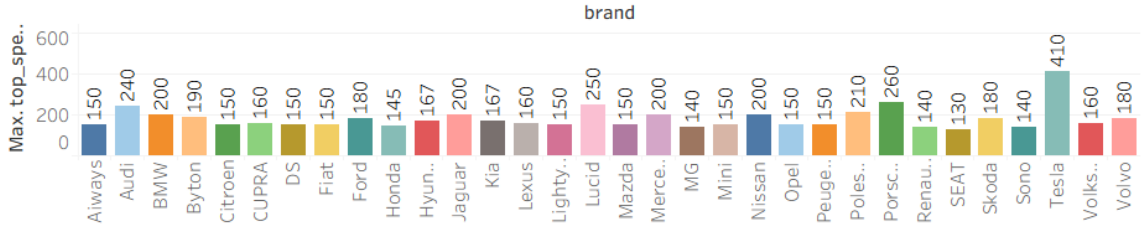


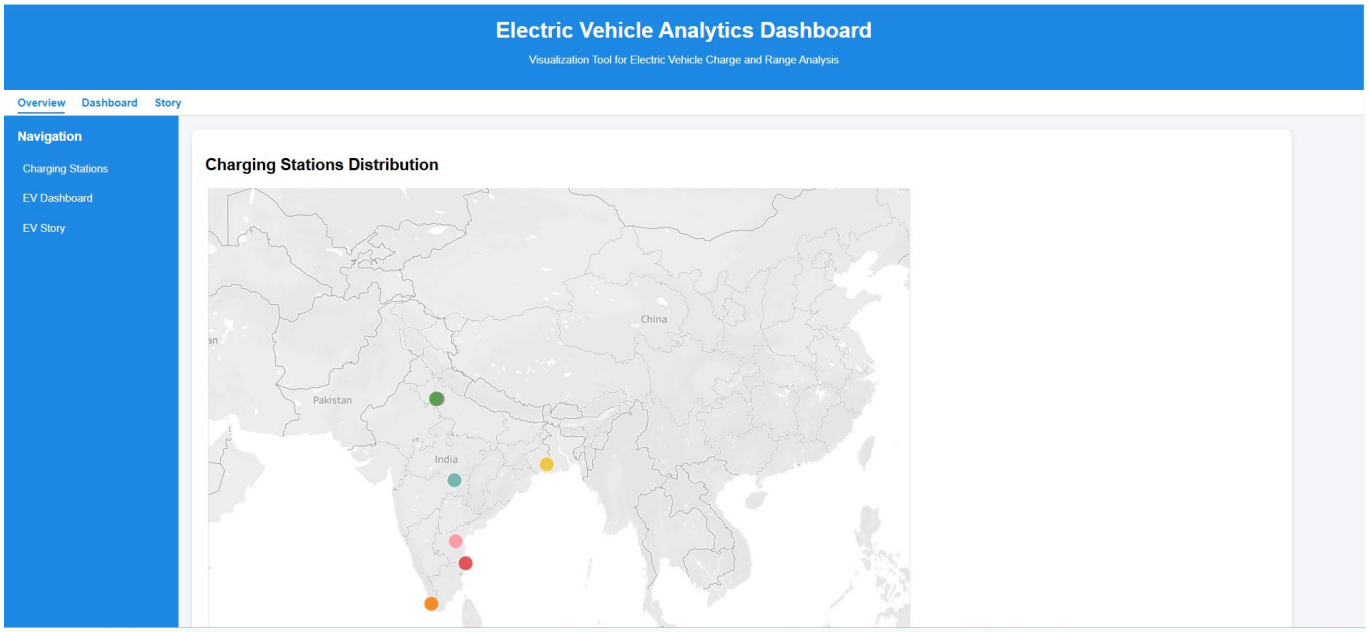


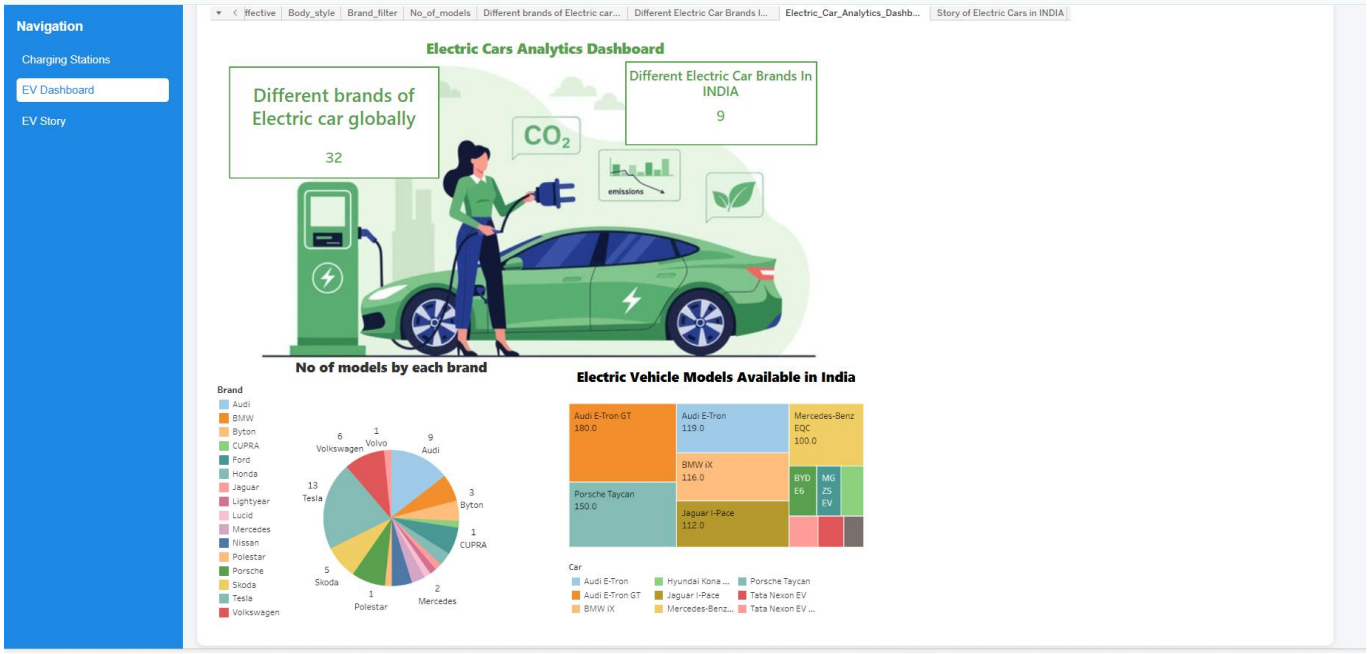
### Maximum Top Speed of Electric Vehicle Brands



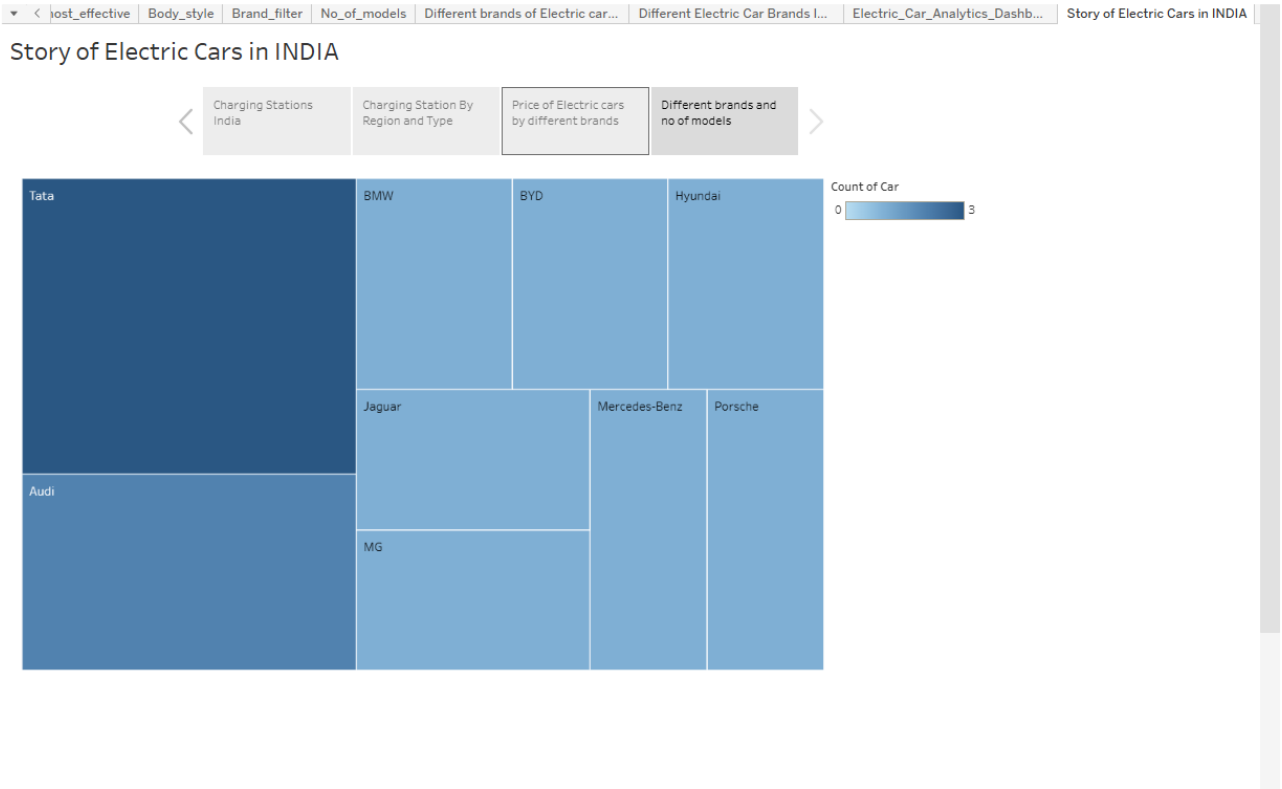
# 9. Web Integration

To improve accessibility, the Tableau dashboards were embedded into a web application using Flask. Flask is a lightweight Python web framework that allows developers to build simple web interfaces quickly. Using iframe integration, Tableau dashboards and stories were embedded directly into the website so users can interact with visualizations in a custom-built interface.



## Story of Electric Cars in India

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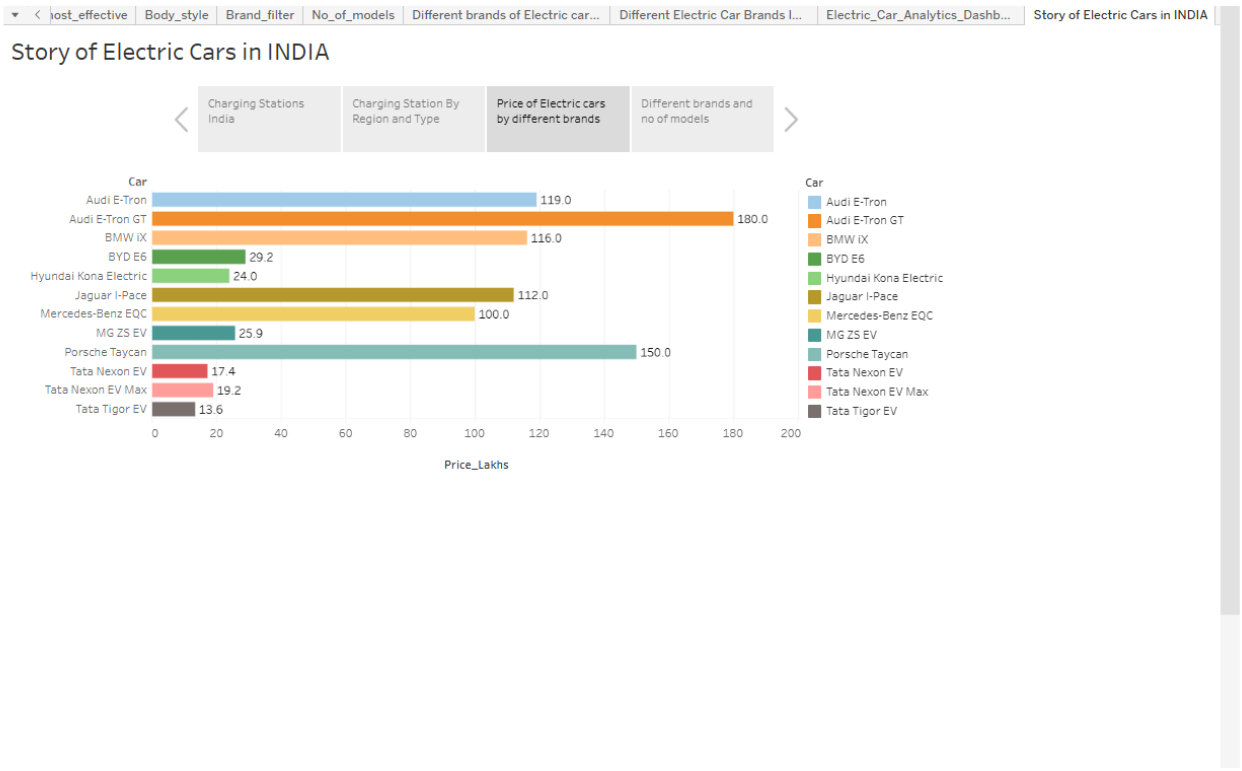

# 10. Results and Insights

The analysis revealed several important insights:

- Tesla and Audi offer a large number of EV models compared to other brands.
- SUVs dominate the EV market in terms of body style.
- Tesla shows the highest top speed among EV brands.
- Premium EV models such as Porsche Taycan and Audi E-Tron GT have significantly higher prices.
- Charging infrastructure is concentrated in certain regions, indicating uneven distribution across areas.

# 11. Conclusion

This project demonstrates how EV datasets can be analyzed using data visualization techniques to uncover useful insights. By cleaning multiple datasets and combining them into dashboards, meaningful patterns related to EV pricing, efficiency, and infrastructure were identified. The project also shows how dashboards can be embedded into web applications to improve accessibility. Future work may include analyzing larger datasets and incorporating real-time EV charging station data for deeper analysis.